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A modified forward and backward reaching inverse kinematics based incremental control for space manipulators

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Abstract Forward and backward reaching inverse kinematics (FABRIK) is an efficient two-stage iterative solver for inverse kinematics of spherical-joint manipulator without the calculation of Jacobian matrix. Based on FABRIK, this paper presents an incremental control scheme for a free-floating space manipulator consists of revolute joints and rigid links with the consideration of joint constraints and dynamic coupling effect. Due to the characteristics of FABRIK, it can induce large angular movements on specific joints. Apart from that, FABRIK maps three dimensional (3D) problem into two dimensional (2D) problem by a simple geometric projection. This operation can cause infinite loops in some cases. In order to overcome these issues and apply FABRIK on space manipulators, an increments allocation method is developed to constrain the angular movements as well as to re-orient the end-effector. The manipulator is re-positioned based on the momentum conservation law. Instead of pure target position tracking, the orientation control of the end-effector is also considered. Numerical simulation is performed to testify and demonstrate the effectiveness and reliability of the proposed incremental control approach.

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1. Introduction

Due to the growing application demands for automation, dexterity and efficiency in On Orbit Servicing (OOS) and Active

Debris Removal (ADR) missions, robotic capture by space manipulator has been considered as one of the most promising techniques.^{1–3} Generally, a space manipulator is mounted on certain “floating base”, such as satellite, spacecraft, space station, etc. According to the momentum conservation law, the motion of the manipulator varies the position and orientation of the floating base, and the translation and rotation of the base in turn affects the motion of the manipulator as well. This kind of phenomenon is called dynamic coupling.⁴

The effect of dynamic coupling can be neglected in cases where the mass and inertia of the space manipulator are much smaller than the base, or the position and orientation of the

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base are fully controlled. In such cases, the space manipulators are usually approximated as fixed base manipulators, similar to those on the ground.^{5,6} During the past several decades, plenty of work has been reported in the literature dealing with the target estimation, motion planning and control issues of fixed base manipulators, which can be easily transplanted to tackle similar problems of space manipulators when the fixed-base approximation holds. It is worth mentioning that the fixed-base approximation may degrade the control accuracy and the dynamic performance.

The control of space manipulators will be complicated when dynamic coupling is not negligible. Generally, the control objective of a manipulator is a desired position of the end-effector in the Cartesian space, and the control input is the angular position or the angular velocity in joint space.^{7,8} Existing control schemes can be classified into three categories: dynamics-based, data-based and kinematics-based.

Remarkable amount of the control approaches in the literature are dynamics-based. Nevertheless, in dynamics-based approaches, many parameters need to be measured and fed back in practice, such as angular position and velocity, force/torque etc. Besides, model uncertainties and external disturbances are usually troublesome to cope.⁹ With the consideration of dynamic coupling, Vafa and Dubowsky¹⁰ introduced an analytical modeling method for space manipulators called the virtual manipulator (VM), which could be considered as an ideal kinematic chain with a fixed virtual base. Liang et al.¹¹ proposed a practical solution to map a floating base manipulator to a fixed base manipulator, named dynamically equivalent manipulator (DEM), with preserving both dynamic and kinematic properties. Both VM and DEM involve a passive spherical joint, which is not controllable. Based on the momentum conservation law, Umetani and Yoshida¹² derived a new Jacobian matrix for space manipulators called the generalized Jacobian matrix (GJM), which enabled the extension of a variety of control methods developed for a fixed base manipulator to a space manipulator by simply replacing the traditional Jacobian matrix into GJM. Due to the development of artificial intelligence technology, data-based approaches have emerged in recent years. The advantage of data-based approaches is that they do not require the dynamic and/or kinematic model of the manipulator, which is generally favored, especially when the model is hard to derive or highly nonlinear. But data-based approaches require large a priori data set to ensure the training accuracy of the model, which may not be available in space applications.

Intuitively, it is more convenient to control a space manipulator by joint angles, like in those kinematics-based approaches.^{13,14} However, the inverse kinematics will be encountered. By adopting pseudo-inverse of the Jacobian matrix and considering angular velocity limits as constraints, a predictive kinematic control scheme and an incremental kinematic control scheme were reported respectively, where the Jacobian matrix should be non-singular.^{15,16} Yang et al.¹⁷ transformed the motion control of a dual-arm space manipulator to an optimization problem as well. With joint acceleration considered as the control input, the end-effector could achieve the predefined waypoints without consideration of inverse kinematics. Xu et al.¹⁸ proposed a hybrid modeling and analysis method to deal with the dynamic coupling. By resolving the linear and angular momentum conservation equations in different ways, the calculation of Jacobian matrix and its

inverse was avoided. Aristidou and Lasenby¹⁹ developed an efficient iterative solver for inverse kinematics in frame animation production, called forward and backward reaching inverse kinematics (FABRIK), where the Jacobian matrix was avoided. The FABRIK can easily incorporate with joint constraints, and is appropriate for multiple chains with multiple end-effectors. However, it tends to induce large angular movements on specific joints due to the characteristics of FABRIK, which is subject to the configuration of manipulator as well as the desired position and orientation of the end-effector. Furthermore, the FABRIK maps 3D problem into 2D problem by a simple geometric projection, this may cause infinite loops in specific cases.

In order to overcome the abovementioned issues and adaptively applying the FABRIK to address the control problem of a space manipulator, an angular increments allocation method is developed in this paper to break the infinite loops as well as to re-allocate the angular movements. Moreover, instead of pure target position tracking in the FABRIK, the orientation control of the end-effector is also considered.

The remainder of this paper is organized as follows. In Section 2, the basic procedures of the FABRIK in two dimensional case and three dimensional case are formulated, and the problems of applying FABRIK on space robotic manipulators are pointed out. A modified FABRIK-based incremental control approach is described in Section 3. In order to validate the proposed approach, numerical simulation results are reported and analyzed in Section 4. The conclusion of the paper is summarized in Section 5.

2. Preliminary

The FABRIK is an iterative solution for inverse kinematics. As the name implies, the basic procedure of one complete FABRIK iteration can be divided into two stages: the forward (inward) reaching stage and the backward (outward) reaching stage.¹⁹ A schematic diagram of the FABRIK for a two dimensional (2D) chain with fixed base connected by 1 DOF (degree of freedom) revolute joints is shown in Fig. 1, where Fig. 1(a) to Fig. 1(c) describe the forward reaching stage, and Fig. 1(d) to Fig. 1(f) describe the backward reaching stage.

In order to formulate the FABRIK, we assume that the inertia coordinate frame is fixed in space. With respect to the inertia frame, the actual position of each joints, as well as the actual and desired position of the end-effector are denoted as $\mathbf{J}_i (i = 1, 2, \dots, n)$, $\mathbf{X}_e$ and $\mathbf{X}_{ed}$, respectively. Each of the link length of the manipulator is denoted as $l_i (i = 1, 2, \dots, n)$, as shown in Fig. 1. To avoid notation abuse and keep the dimension of variables consistent, the position vectors and matrices mentioned in this paper are all extended into homogeneous formation. For instances, a two dimensional vector can be extended as $\mathbf{J}_i = [x_i, y_i, 0, 1]^T$, and a three dimensional vector can be extended as $\mathbf{J}_i = [x_i, y_i, z_i, 1]^T$.

Based on the above definitions, the forward reaching stage estimates each of the joint position starting from the end-effector to the root joint by the following steps:

- Let the position of the end-effector overlap with $\mathbf{X}_{ed}$, denoted by $\mathbf{X}'_e$, such that $\mathbf{X}'_e = \mathbf{X}_{ed}$;
- Find the new position of the n^{th} joint $\mathbf{J}'_n$ on the line formed by $\mathbf{J}_n$ and $\mathbf{X}'_e$, and satisfying $|\mathbf{J}'_n - \mathbf{X}'_e| = l_n$;

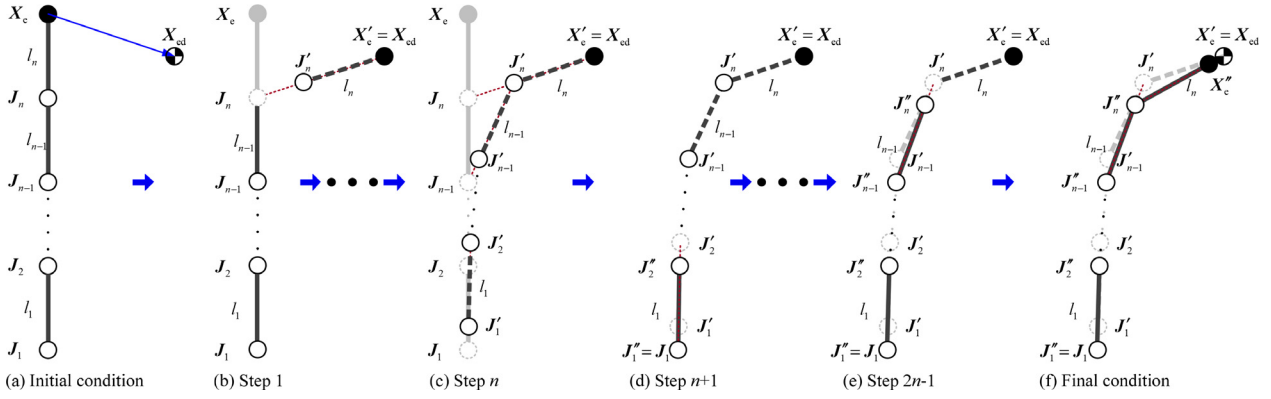


Fig. 1 A schematic diagram of one complete iteration of FABRIK in 2D.

- (c) Find the new position of the i^{th} joint J_i on the line formed by J_i and J'_{i+1} , and satisfying $|J'_i - J'_{i+1}| = l_i$;

Therefore, the steps of the forward reaching stage can be concisely formulated as

$$\begin{cases} X'_e = X_{ed} \\ J'_n = X'_e + l_n(J_n - X'_e)/|J_n - X'_e| \\ J'_i = J'_{i+1} + l_i(J_i - J'_{i+1})/|J_i - J'_{i+1}| \quad i = 1, 2, \dots, n-1 \end{cases} \quad (1)$$

The backward reaching stage re-evaluates each of the joint position starting from the root joint to the end-effector. In order to distinguish from forward reaching stage, the position of each joint as well as the end-effector are with a two-prime superscript. The procedure of the backward reaching stage is described by the following steps:

- (d) Because the base is fixed, the root joint position in backward reaching stage should satisfy $J''_1 = J_1$;
(e) Let J''_{i+1} be on the line formed by J''_i and J'_{i+1} , and satisfying $|J''_i - J''_{i+1}| = l_i$;
(f) Let X''_e be on the line formed by X'_e and J''_n , and satisfying $|J''_n - X''_e| = l_n$.

Consequently, the steps of the backward reaching stage can be formulated as

$$\begin{cases} J''_1 = J_1 \\ J''_{i+1} = J''_i + l_i(J'_{i+1} - J''_i)/|J'_{i+1} - J''_i| \quad i = 1, 2, \dots, n-1 \\ X''_e = J''_n + l_n(X'_e - J''_n)/|X'_e - J''_n| \end{cases} \quad (2)$$

After one iteration of steps (a) to (f), the procedure is repeated until the desired position of the end-effector is achieved. From the above procedure, it is noted that the FABRIK is prone to induce large angular movements on specific joints, such as those close to the end-effector in Fig. 1.

For three dimensional (3D) cases, the procedure described above can be directly adopted if a chain is connected by ideal ball-and-socket joints (3 DOFs). Considering the fact that robotic manipulators are connected mostly by revolute joints to achieve 3D motion of the end-effector in practical applications, the forward reaching stage is remaining the same as Eq. (1), and the re-evaluation procedure of a revolute joint position with joint restriction in the backward reaching stage is

depicted in Fig. 2(a). Because J''_{i+1} should be within the dash line plane, in order to determine J''_{i+1} , the position of J'_{i+1} is replaced by its projection onto this plane, denoted by $\tilde{J}'_{i+1}$.

The DH frames of a multi-joint robotic manipulator are defined accordingly,²⁰ and let T_i denote the homogenous transformation matrix from the inertia frame to the i^{th} DH frame. The elements of T_i are trigonometric functions of joint variables $q_1, q_2, \dots, q_i$ and other DH parameters. Then according to the DH convention, T_i could be easily obtained. Further define the projection matrix as

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

Calculate the projection of J'_{i+1} by

$$\tilde{J}'_{i+1} = J'_i + T_i^{-1} P T_i (J'_{i+1} - J'_i) \quad (4)$$

Then find the position J''_{i+1} on the line formed by $\tilde{J}'_{i+1}$ and J''_i , and satisfying $|J''_i - J''_{i+1}| = l_i$. Thus, the steps of the backward reaching stage in 3D cases can be formulated as

$$\begin{cases} J''_1 = J_1 \\ \tilde{J}'_{i+1} = J'_i + T_i^{-1} P T_i (J'_{i+1} - J'_i) \quad i = 1, 2, \dots, n-1 \\ J''_{i+1} = J''_i + l_i(\tilde{J}'_{i+1} - J''_i)/|\tilde{J}'_{i+1} - J''_i| \quad i = 1, 2, \dots, n-1 \\ X''_e = J''_n + l_n(X'_e - J''_n)/|X'_e - J''_n| \end{cases} \quad (5)$$

However, in specific cases where $\tilde{J}'_{i+1}$ falls on the line formed by J'_{i+1} and J''_i , as shown in Fig. 2(b), J''_{i+1} overlaps

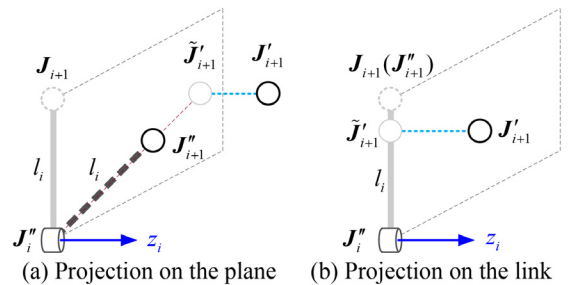


Fig. 2 Geometric projection in backward reaching stage.

with $\mathbf{J}_{i+1}$. Consequently, infinite loops issue arises.

3. FABRIK-based incremental control

In order to adopt FABRIK to solve the position tracking control problem of a space manipulator, the forward reaching stage is remaining the same as described in Section 2. Considering the fact that the robotic manipulator is assumed as spherical-joint (3 DOFs) type in the forward reaching stage, and the end-effector is oriented to the desired position with the consideration of robotic geometry only. Subsequently, the angular movements as well as the robotic configuration may be impossible to be achieved by the revolute-joint (1 DOF) space robot. Therefore, in the backward reaching stage, the end-effector is re-oriented by applying physical constraints to limit the joint velocity, and considering the rotational degree of freedom of the revolute-joint to update the robotic configuration.

In the backward reaching stage, the position of the base do not need to keep because of the free floating nature. Therefore, the position of the root joint is set to maintain the estimated position in the forward reaching stage. In order to distinguish notations from Eq. (5), it is written as

$$\bar{\mathbf{J}}''_1 = \mathbf{J}'_1 \quad (6)$$

In order to constrain the angular movements, joint velocity limits are applied while re-evaluating the next joint position. First, according to the geometric projection method illustrated in Fig. 2, find the position of the $(i+1)$ th joint, denoted as $\bar{\mathbf{J}}''_{i+1}$, such that

$$\begin{cases} \bar{\mathbf{J}}_{i+1} = \bar{\mathbf{J}}''_i + \mathbf{T}_i^{-1} \mathbf{P} \mathbf{T}_i (\mathbf{J}'_{i+1} - \bar{\mathbf{J}}''_i) & i = 1, 2, \dots, n-1 \\ \mathbf{J}''_{i+1} = \mathbf{J}''_i + l_i (\mathbf{J}''_{i+1} - \mathbf{J}''_i) / |\mathbf{J}''_{i+1} - \mathbf{J}''_i| & i = 1, 2, \dots, n-1 \\ \mathbf{X}''_e = \mathbf{J}''_n + l_n (\mathbf{X}''_e - \mathbf{J}''_n) / |\mathbf{X}''_e - \mathbf{J}''_n| \end{cases} \quad (7)$$

Define the dimensionality reduction matrix as

$$\mathbf{E}_{2d} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (8)$$

Such that the movements of each joint position with respect to the revolution plane of the previous joint can be determined by

$$(\Delta \bar{x}_i, \Delta \bar{y}_i)^T = \mathbf{E}_{2d} \mathbf{T}_i'' (\bar{\mathbf{J}}''_{i+1} - \mathbf{J}''_i) \quad (9)$$

In the backward reaching stage, the joint variable will be updated by $q_i'' = q_i + \Delta q_i$, where Δq_i denotes an incremental joint angle that within the i th joint constraint. Therefore, the transformation matrix is represented as $\mathbf{T}_i''$, whose elements are trigonometric functions of $q_1'', q_2'', \dots, q_i''$ and other DH parameters.

If x_i , the x -axis of the i th DH frame, is parallel with link l_i and pointing to $\mathbf{J}_{i+1}$, as shown in Fig. 3(a), the incremental joint angle is calculated by

$$\Delta \bar{q}_i = \arctan(\Delta \bar{y}_i / \Delta \bar{x}_i) \quad (10)$$

If y_i , the y -axis of the i th DH frame, is parallel with link l_i and pointing to $\mathbf{J}_{i+1}$, as shown in Fig. 3(b), the incremental joint angle should be calculated by

$$\Delta \bar{q}_i = \arctan(\Delta \bar{x}_i / \Delta \bar{y}_i) \quad (11)$$

If z_i , the z -axis of the i th DH frame, is parallel with link l_i and pointing to $\mathbf{J}_{i+1}$, the joint angle is set the same as the previous one, that is

$$\Delta \bar{q}_i = 0 \quad (12)$$

Assume the joint velocity limits are $\dot{q}_{1\max}, \dot{q}_{2\max}, \dots, \dot{q}_{n\max}$, and denote the control period as t_p , then the following relationship should be keep, such that

$$|\Delta q_i| \leq \dot{q}_{i\max} t_p \quad i = 1, 2, \dots, n \quad (13)$$

However, the calculated incremental joint angles in Eq. (10) or Eq. (11) may not satisfy the inequality in Eq. (13) all the time. In order to constrain the angular movements, if $|\Delta \bar{q}_i| \leq \dot{q}_{i\max} t_p$, as illustrated in Fig. 4(a), the incremental joint angle and joint position as well as the position of the end-effector are updated by

$$\begin{cases} \Delta q_i = \Delta \bar{q}_i \\ \mathbf{J}''_{i+1} = \bar{\mathbf{J}}''_{i+1} \\ \mathbf{X}''_e = \bar{\mathbf{X}}''_e \end{cases} \quad (14)$$

Otherwise if $|\Delta \bar{q}_i| > \dot{q}_{i\max} t_p$ as illustrated in Fig. 4(b), update the incremental joint angle by

$$\Delta q_i = \text{sgn}(\Delta \bar{q}_i) \dot{q}_{i\max} t_p \quad (15)$$

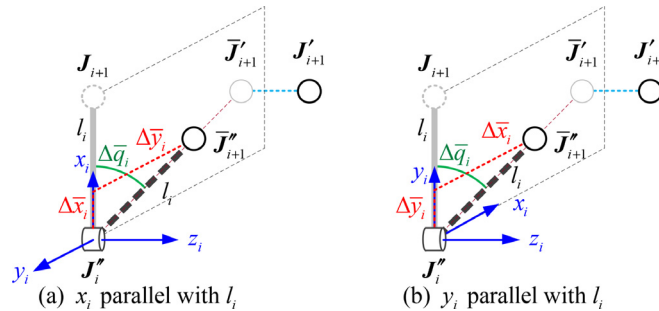


Fig. 3 Projection in backward reaching stage with consideration of joint restriction.

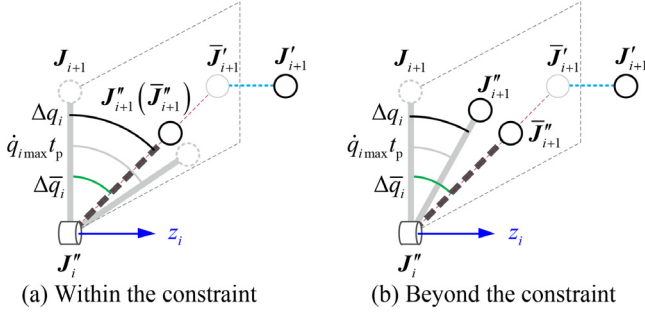


Fig. 4 Process of constraining angular movement.

Subsequently, the Cartesian coordinates of the joint with respect to the inertia frame are determined by

$$\begin{cases} \mathbf{J}''_{i+1} = \mathbf{J}''_i + (\mathbf{T}''_i)^{-1} \mathbf{L}_i \\ \mathbf{X}''_e = \mathbf{J}''_n + (\mathbf{T}''_n)^{-1} \mathbf{L}_n \end{cases} \quad (16)$$

where $\mathbf{L}_i = [l_i \sin(\Delta q_i), l_i \cos(\Delta q_i), 0, 0]^T$ or $\mathbf{L}_i = [l_i \cos(\Delta q_i), l_i \sin(\Delta q_i), 0, 0]^T$ or $\mathbf{L}_i = [0, 0, l_i, 0]^T$. Similar to Eq. (10), Eq. (11) and Eq. (12), $\mathbf{L}_i$ is subject to the direction of the axes of the i th DH frame.

The process described by Eqs. (6)–(16) indicate the modified backward reaching stage with the consideration of joint constraints. Consequently, the end-effector is reoriented, meanwhile, the joint velocity limits are applied to constrain the angular movements.

For a free floating space manipulator, as illustrated in Fig. 5, the center of mass can be calculated by

$$\mathbf{r}_g = \sum_{i=0}^n (m_i \mathbf{r}_i) / \sum_{i=0}^n m_i \quad (17)$$

where m_i is the mass of each link (or base), $\mathbf{r}_i$ is the vector pointing from the origin of the inertia frame to the center of mass of each link (or base) $\mathbf{C}_i$, and $\mathbf{r}_g$ denotes the vector pointing from the origin of the inertia frame to the center of mass of the space manipulator. To keep dimensional consistence, $\mathbf{r}_i$ and $\mathbf{r}_g$ are also extended to homogeneous formation.

After the new backward reaching stage, the center of mass of the space manipulator is

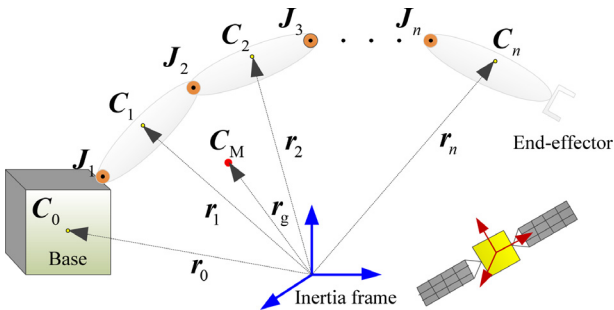


Fig. 5 An illustration of a free floating space manipulator.

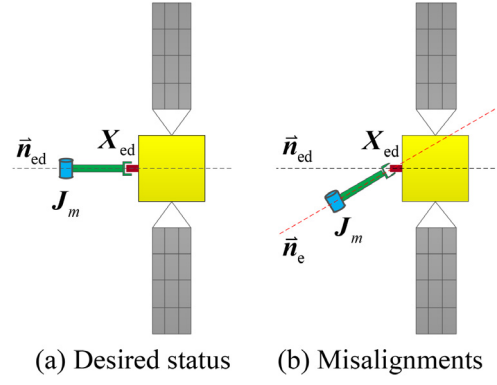


Fig. 6 Robotic capture scenarios.

$$\mathbf{r}''_g = \sum_{i=0}^n (m_i \mathbf{r}''_i) / \sum_{i=0}^n m_i \quad (18)$$

However, according to the momentum conservation law, the center of mass of a free floating space manipulator should be fixed because no external force exerts on the manipulator system. Thus, the obtained joint position $\mathbf{J}'_1, \mathbf{J}'_2, \dots, \mathbf{J}'_n$ should be further corrected by

$$\begin{cases} \tilde{\mathbf{J}}''_i = \mathbf{J}''_i - (\mathbf{r}''_g - \mathbf{r}_g) & i = 1, 2, \dots, n \\ \tilde{\mathbf{X}}''_e = \mathbf{X}''_e - (\mathbf{r}''_g - \mathbf{r}_g) \end{cases} \quad (19)$$

Because the space manipulator is re-oriented in the modified backward reaching stage, the infinite loop situation is changed after each of the modified FABRIK iteration. Once the situation changed, $\tilde{\mathbf{J}}_{i+1}$ will not fall on the line formed by $\mathbf{J}_{i+1}$ and $\mathbf{J}'_i$ in the next iteration. As a result, the infinite loop will be broken. By iteratively executing the forward and new backward reaching process, the desired position of the end-effector could be achieved eventually.

To perform robotic capture, not only the desired position should be achieved, but also the end-effector should approach the desired position in specific direction as shown in Fig. 6(a). If misalignments exist, as shown in Fig. 6(b), capture failure will most likely occur, and may cause serious consequences.

Without loss of generality, the desired orientation of the end-effector is denoted as $\vec{\mathbf{n}}_{ed} = [n_x, n_y, n_z, 1]^T$. Since the end-effector is fixed to the last link, the nearest joint should achieve a desired position in order to satisfy both position and orientation requirements of the end-effector. It is worth mentioning that the revolute axis of the joint, denoted by $\mathbf{J}_m$, should not be parallel with the last link. The desired position of this joint is

$$\mathbf{J}_{md} = \mathbf{X}_{ed} - |\mathbf{X}_e - \mathbf{J}_m| \vec{\mathbf{n}}_{ed} / |\vec{\mathbf{n}}_{ed}| \quad (20)$$

It is noted that the desired position of the joint is derived by both the desired position and orientation of the end-effector. Therefore, once the desired position of the last joint is achieved, both the desired position and orientation of the end-effector could be achieved. To conclude the modified FABRIK-based incremental control, the pseudo code of the iterative control procedure is presented as

Algorithm: Modified FABRIK-based incremental control

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1 given the initial configuration of the manipulator
    $J_1, J_2, \dots, J_n, q_1, q_2, \dots, q_n$ ;
2 given the joint velocity limits  $\dot{q}_{1\max}, \dot{q}_{2\max}, \dots, \dot{q}_{n\max}$ ;
3 calculate the mass center of the manipulator  $r_g$ ;
4 evaluate the transformation matrix of DH frames
    $T_1, T_2, \dots, T_n$ ;
5 given the desired position and orientation of the end-
   effector  $X_{ed}, \bar{n}_{ed}$ ;
6 calculate the desired position of last joint
    $J_{md} = X_{ed} - |X_e - J_m| \bar{n}_{ed} / |\bar{n}_{ed}|$ ;
7 while  $|J_{md} - J_m| > \text{tolerance}$  do
8   forward reaching stage
9   for each joint  $J_i (i = n, n-1, \dots, 2, 1)$  do
10     $X'_e = X_{ed}$ ;
11     $J'_m = J_{md}$ ;
12     $J'_i = J_{i+1} + l_i(J_i - J_{i+1}) / |J_i - J_{i+1}|$ ;
13   backward reaching stage
14   for each joint  $J_i (i = 1, 2, \dots, n)$  do
15     $J'_1 = J'_1$ ;
16     $J'_{i+1} = J'_i + T_i^{-1} P T_i (J'_{i+1} - J'_i)$ ;
17     $J'_{i+1} = J'_i + l_i (J'_{i+1} - J'_i) / |J'_{i+1} - J'_i|$ ;
18     $X''_e = J'_n + l_n (X'_e - J'_n) / |X'_e - J'_n|$ ;
19     $(\Delta \bar{x}_i, \Delta \bar{y}_i)^T = E_{2d} T'_i (J'_{i+1} - J'_i)$ ;
20     $\Delta \bar{q}_i = \arctan(\Delta \bar{x}_i / \Delta \bar{y}_i)$  or  $\Delta \bar{q}_i = \arctan(\Delta \bar{y}_i / \Delta \bar{x}_i)$ ;
21    if  $|\Delta \bar{q}_i| \leq q_{i\max} t_p$  do
22      $\Delta q_i = \Delta \bar{q}_i$ ;
23      $J'_{i+1} = J'_{i+1}$ ;
24      $X''_e = X'_e$ ;
25    else do
26      $\Delta q_i = \text{sgn}(\Delta \bar{q}_i) q_{i\max} t_p$ ;
27      $J'_{i+1} = J'_i + (T'_i)^{-1} L_i$ ;
28      $X''_e = J'_n + (T'_n)^{-1} L_n$ ;
29    update the joint angle  $q'_i = q_i + \Delta q_i$ ;
30    re-evaluate  $T'_{i+1}$ ;
31   calculate the new mass center of the manipulator  $r''_g$ ;
32   for each joint  $J'_i (i = 1, 2, \dots, n)$  do
33    translate the joint position  $J'_i = J'_i - (r''_g - r_g)$ ;
34     $X''_e = X''_e - (r''_g - r_g)$ ;
35   update the joint position  $J_i = J'_i, X_e = X''_e$ ;
36   output the control variable  $\Delta q_1, \Delta q_2, \dots, \Delta q_n$ ;

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4. Simulation results and analysis

In order to validate the proposed incremental control scheme, numerical simulation is performed on a 6R free floating robotic manipulator. The initial position of the manipulator and possible assignment of DH frames are shown in Fig. 7. The inertia frame $Oxyz$ is fixed in the space.

The prototype of the manipulator is UR5 (universal-robots), and the joint velocity limits are ± 180 (o)/s for all the joints. In the simulation, the manipulator is assumed to be mounted on a 100 kg free floating base satellite, the link length and mass are listed in Table 1. Accordingly, the DH parameters are listed in Table 2, where $q_1, q_2, \dots, q_6$ denote the joint variable.

Within the working space, for given desired position and orientation of the end-effector in homogenous formation, such

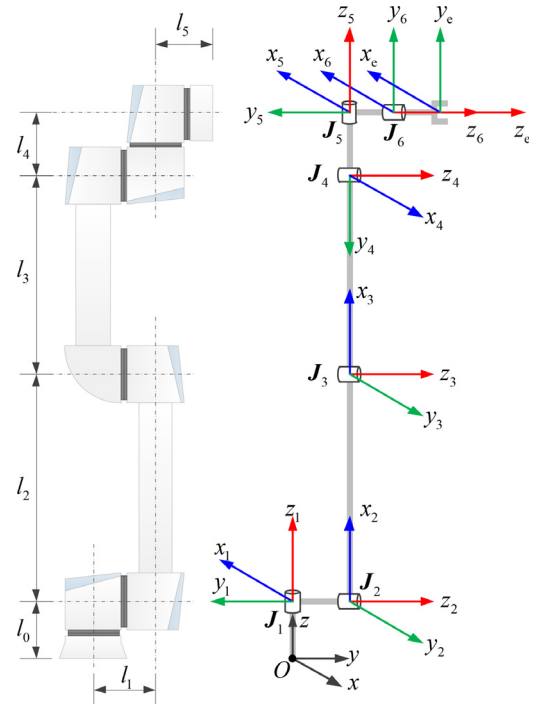


Fig. 7 Initial position of manipulator and assignment of DH frames.

as $X_{ed} = [0.3 \text{ m}, 0.5 \text{ m}, 0.8 \text{ m}, 1]^T$ and $\bar{n}_{ed} = [1, 1, -1, 0]^T$, set the control period as $t_p = 0.01 \text{ s}$ and tolerance as 0.01 m , the simulation results are shown in Figs. 8–10.

Fig. 8 intuitively shows the configuration of the manipulator at typical time steps. Starting from the home position, the end-effector moves towards the desired position as expected. Simultaneously, joint 5 approaches its desired position in order to achieve both the desired position and orientation of the end-effector.

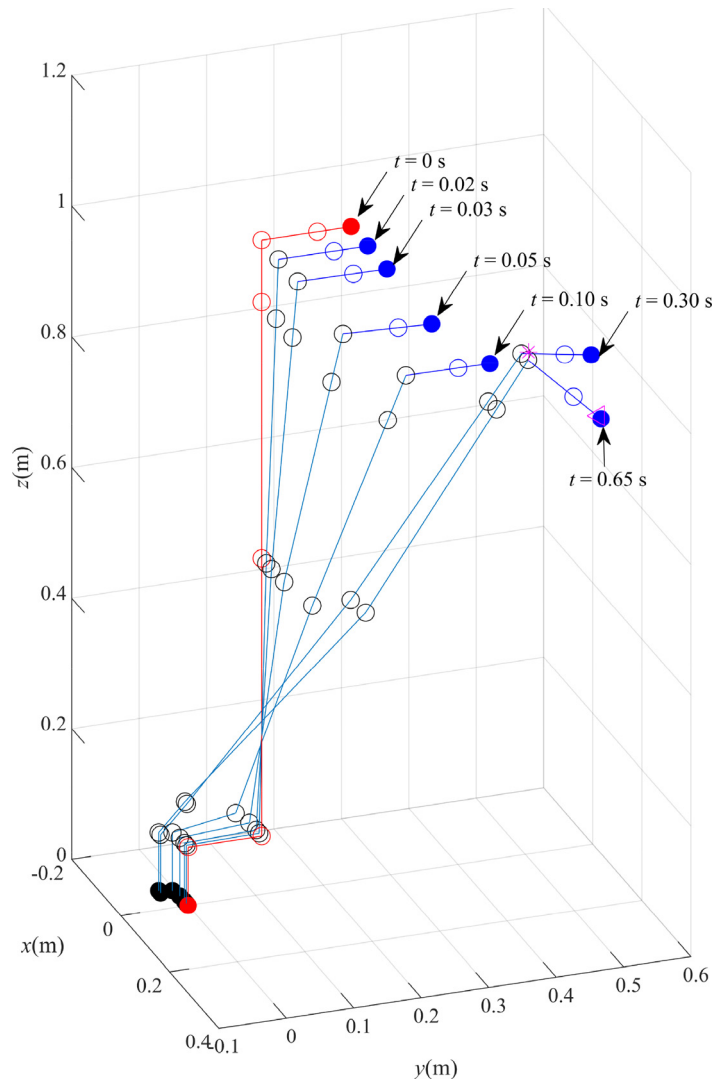
Fig. 9 shows the trajectory of the end-effector. It can be seen that the end-effector achieves the desired position after 0.62 s . Because the z -axis of joint 6 is parallel with link 5 and link 6, as shown in Fig. 7, it has no contribution to the position of the end-effector. The corresponding joint angles are shown in Fig. 10. The large variation of joint angles are mainly caused by the setting of joint velocity limits. Moreover, as formulated in Eq. (20), the desired position and direction of the end-effector are mapped to the desired position of the last revolute joint, where the revolute axis is not parallel with the last link. Then because the revolute axis of joint 6 is parallel with link 5 and link 6, the desired position and direction of the end-effector in the simulation are mapped to the desired position of joint 5. According to the proposed approach, the desired position and direction of the end-effector could be achieved by the first five joints. As a result, joint 6 becomes independent, which could be utilized for adjusting the angular bias between the end-effector and the grasping feature on the target. The above simulation results indicate that the proposed approach is effective in position and orientation control.

Table 1 Length and mass distributions.

Index	0	1	2	3	4	5	6
$l_i(\text{m})$	0.08916	0.10915	0.42500	0.39200	0.09475	0.08250	0.05000
$m_i(\text{kg})$	100	1.4	7.8	6.2	1.4	1.4	0.2

Table 2 DH parameters of space manipulator.

Index	$\alpha_{i-1}(^\circ)$	$a_{i-1}(\text{m})$	$d_i(\text{m})$	$\theta_i(^\circ)$
1	0	0	l_0	$180 + q_1$
2	90	0	l_1	$90 + q_2$
3	0	l_2	0	q_3
4	0	l_3	0	$90 + q_4$
5	90	0	l_4	$180 + q_5$
6	90	0	l_5	q_6

**Fig. 8** Configuration of manipulator at typical time steps.

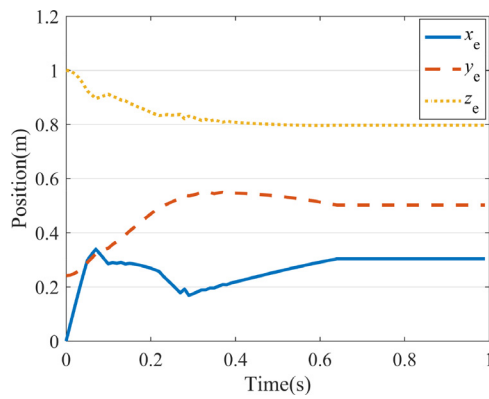


Fig. 9 Trajectory of end-effector.

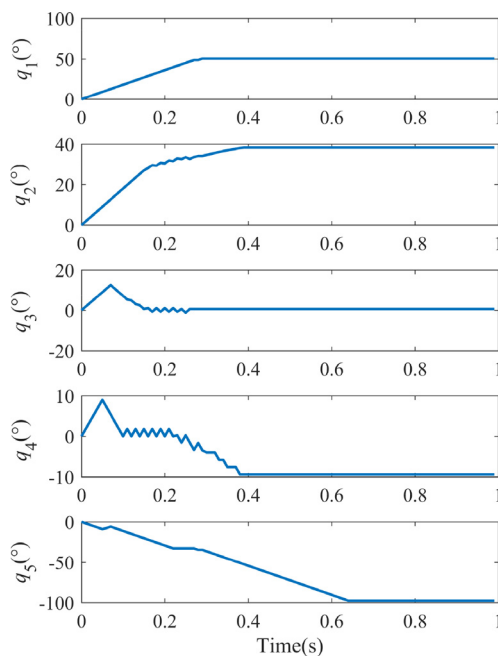


Fig. 10 History of joint angles.

5. Conclusion

This paper presents an incremental control scheme for a free-floating space manipulator based on modified FABRIK. FABRIK is an efficient iterative solver for inverse kinematics of spherical-joint robotic manipulators. But in practical applications, most of the manipulators are consisted of one degree of freedom revolute-joint. In order to achieve both the desired position and orientation of the end-effector, the desired position of the nearest joint to the end-effector is derived, where the revolute axis of the joint is not parallel with the last link. Then the joint position is set as the derived desired one in the procedure of forward reaching stage. During the backward reaching stage, incremental joint angles are obtained ultimately with the consideration of joint velocity limits as well as the dynamic coupling

effect. The advantages of the proposed scheme could be summarized as: (a) the basic FABRIK is modified for solving the inverse kinematics of practical revolute-joint robot; (b) because the end-effector is re-oriented in the modified backward reaching stage, the infinite iteration problem of the basic FABRIK is eliminated; (c) it is a Jacobian-free approach, such that, the complex calculation of matrices and non-singularity assumption of the Jacobian matrix is avoided; (d) Since the control input is incremental joint angle, multiple-solution-problem of robotic inverse kinematics is avoided. Numerical simulation results demonstrate the effectiveness of the proposed control scheme. The calculation of Jacobian matrix and the multiple solution issues are avoided by the proposed approach, which is generally favored in robotic capture and has great potential in practical applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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